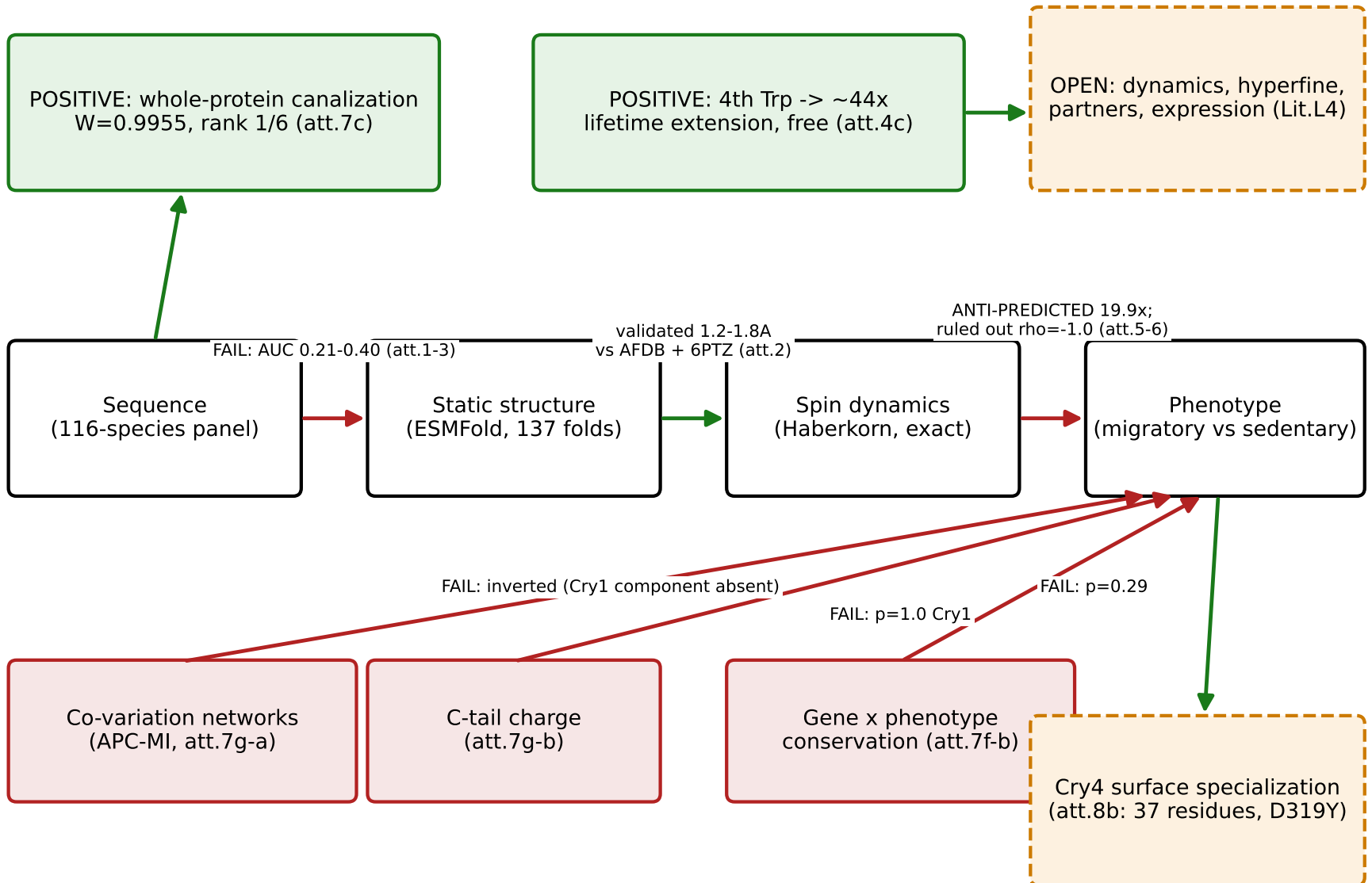


Causal map of the evidence: sequence -> structure -> spin dynamics -> phenotype



Green = supported (attempt nos.); red = falsified under pre-registered gates; orange dashed = open degrees of freedom. Labels transcribe committed results only.